
MODULE *FormalizedCorollary*

EXTENDS *Integers, FiniteSets, Sequences, Utils*

Segments

$Segment(a, b) \triangleq \langle a, b \rangle$

$Seg \triangleq \{s \in Int \times Int : s[1] \leq s[2]\}$

$EmptySeg \triangleq \langle \rangle$ the empty segment is not part of *Seg*

$b(s) \triangleq s[1]$

$e(s) \triangleq s[2]$

$SegShiftLeft(s) \triangleq Segment(b(s) - 1, e(s) - 1)$

$SegTruncateLeft(s) \triangleq$
 IF $b(s) = e(s)$
 THEN $EmptySeg$
 ELSE $Segment(b(s) + 1, e(s))$

$Precedes(s1, s2) \triangleq$
 returns $s1 \leq s2$
 $\wedge b(s1) < b(s2)$
 $\wedge e(s1) < e(s2)$

$SegSubsetEq(s1, s2) \triangleq$
 checks if $s1$ is a subset of $s2$
 $\wedge b(s2) \leq b(s1)$
 $\wedge e(s1) \leq e(s2)$

Multisegments

$Multisegments \triangleq Seq(Seg)$

$SubMultisegment(N, M) \triangleq SubMultiset(N, M)$

$$\text{SameMultisegment}(M, N) \triangleq \text{SameBag}(M, N)$$

$$\text{IndexSet}(M) \triangleq 1 \dots \text{Len}(M)$$

$$\text{Delta}(M) \triangleq M$$

sticking to paper notation (redundant)

$$\begin{aligned} \text{IsLadder}(M) &\triangleq \\ &\wedge M \neq \langle \rangle \\ &\wedge \exists R \in \text{Enumerations}(\text{IndexSet}(M)) : \\ &\quad \forall i \in 1 \dots (\text{Len}(M) - 1) : \\ &\quad \quad \text{Precedes}(M[R[i]], M[R[i + 1]]) \end{aligned}$$

a ladder is nonempty

$$\text{Delta}_-\{R[i]\} \text{ Delta}_-\{R[i + 1]\}$$

Depth

$$\text{Depth}(M) \triangleq$$

LET

$$\begin{aligned} \text{IsDepthChain}(i, d, \text{Chain}) &\triangleq \\ &\wedge \text{Chain}[1] = i \\ &\wedge \forall r \in 1 \dots d : \\ &\quad \text{Precedes}(\\ &\quad \quad \text{Delta}(M)[\text{Chain}[r]], \\ &\quad \quad \text{Delta}(M)[\text{Chain}[r + 1]] \\ &\quad) \end{aligned}$$

the chain starts at index i

each segment precedes the next one

IN

$$\begin{aligned} [i \in \text{IndexSet}(M) \mapsto \\ \quad \text{Max}(\{ \\ \quad \quad d \in 0 \dots (\text{Len}(M) - 1) : \\ \quad \quad \quad \exists \text{Chain} \in [1 \dots (d + 1) \rightarrow \text{IndexSet}(M)] : \\ \quad \quad \quad \quad \text{IsDepthChain}(i, d, \text{Chain}) \\ \quad \quad \}) \\ \quad] \end{aligned}$$

there exists a chain of length j from i

$$d(M) \triangleq$$

returns the depth of the multisegment M

IF $M = \langle \rangle$

THEN 0

ELSE $\text{Max}(\text{Range}(\text{Depth}(M)))$

The Map K

$$\begin{aligned} \text{DepthFiber}(M, k) &\triangleq \\ \text{Fiber}(\text{Depth}(M), k) \end{aligned}$$

$d_M^{\wedge\{-1\}}(k)$

$$\text{AdmissibleEnumeration}(M, k) \triangleq$$

LET

$$F \triangleq \text{DepthFiber}(M, k)$$

depth fiber at k

$$\text{IsAdmissible}(P) \triangleq$$

$$\forall r \in 1 \dots (\text{Cardinality}(F) - 1) :$$

$$\begin{aligned} &\text{SegSubsetEq}(\\ &\quad \text{Delta}(M)[P[r+1]], \\ &\quad \text{Delta}(M)[P[r]] \\ &) \end{aligned}$$

P satisfies the admissibility condition

IN

$$\text{CHOOSE } P \in \text{Enumerations}(F) :$$

$$\text{IsAdmissible}(P)$$

choose any admissible enumeration

$$j\text{Index}(M, k) \triangleq$$

$$\text{IF } k \in \text{Range}(\text{Depth}(M))$$

$$\text{THEN } \text{Last}(\text{AdmissibleEnumeration}(M, k))$$

$$\text{ELSE } 0$$

$$\text{IVee}(M) \triangleq$$

LET

$$\text{Cycles} \triangleq$$

$$\{ \text{AdmissibleEnumeration}(M, k) : \\ k \in \text{Range}(\text{Depth}(M)) \}$$

one cycle for each depth fiber

IN

$$\text{PermutationFromCycles}(\text{IndexSet}(M), \text{Cycles})$$

permutation $i \mapsto i^{\wedge\text{vee}}$ induced by these cycles

2.1a

$$\text{SegmentTransform}(M, i) \triangleq$$

this is $_i$

$$\begin{aligned} &\text{Segment}(\\ &\quad b(\text{Delta}(M)[i]), \\ &\quad e(\text{Delta}(M)[\text{IVee}(M)[i]]) \\ &) \end{aligned}$$

2.1b

$$\text{DistinguishedIndices}(M) \triangleq$$

$$\{ j\text{Index}(M, k) : k \in \text{Range}(\text{Depth}(M)) \}$$

$$\text{RemainingIndices}(M) \triangleq$$

this is $\text{I}'' = \text{I} \setminus \text{I}'$

$$IndexSet(M) \setminus DistinguishedIndices(M)$$

$$\begin{array}{l} \text{2.1c} \\ l(M) \triangleq \\ \quad [r \in 1 \dots (d(M) + 1) \mapsto \\ \quad \quad SegmentTransform(M, jIndex(M, r - 1)) \\ \quad] \end{array}$$

$$\begin{array}{l} DerivedMultisegment(M) \triangleq \\ \quad EnumerateValues(\\ \quad \quad [i \in RemainingIndices(M) \mapsto \\ \quad \quad \quad SegmentTransform(M, i)] \\ \quad) \end{array}$$

$$m' = \text{sum of } Delta_i \text{ for } i \text{ in } \Gamma'$$

$$\begin{array}{l} \text{2.2} \\ bTransform(M, i) \triangleq \\ \quad b(SegmentTransform(M, i)) \\ \\ eTransform(M, i) \triangleq \\ \quad e(SegmentTransform(M, i)) \end{array}$$

$$K(M) \triangleq \langle l(M), DerivedMultisegment(M) \rangle$$

The single *MW* step

$$\begin{array}{l} min(M) \triangleq \\ \quad Min(\{b(Delta(M)[i]) : i \in IndexSet(M)\}) \end{array}$$

$$\begin{array}{l} \text{3.1a} - \text{3.1c} \\ LeadingSequence(M) \triangleq \\ \quad LET \\ \quad \quad CandidateSequences \triangleq \\ \quad \quad \quad UNION \{ \\ \quad \quad \quad \quad [1 \dots k \rightarrow IndexSet(M)] : \\ \quad \quad \quad \quad k \in 1 \dots Len(M) \\ \quad \quad \quad \} \end{array}$$

all possible nonempty index sequences of relevant length

$$\begin{array}{l} FirstOK(i) \triangleq \\ \quad \wedge \quad b(Delta(M)[i]) = min(M) \\ \quad \wedge \quad \forall q \in IndexSet(M) : \\ \quad \quad b(Delta(M)[q]) = min(M) \\ \quad \Rightarrow \\ \quad e(Delta(M)[i]) \leq e(Delta(M)[q]) \end{array}$$

$Delta_i$ begins at $min\ M$

$Delta_i$ has minimal ending among these segments

$ \begin{aligned} & \text{NextCandidate}(i, q) \triangleq \\ & \quad \wedge \text{Precedes}(\text{Delta}(M)[i], \text{Delta}(M)[q]) \\ & \quad \wedge b(\text{Delta}(M)[q]) = b(\text{Delta}(M)[i]) + 1 \end{aligned} $	$ \begin{aligned} & \text{Delta_i} \quad \text{Delta_q} \\ & \text{Delta_q begins exactly one later} \end{aligned} $
$ \begin{aligned} & \text{NextOK}(i, \text{next}) \triangleq \\ & \quad \wedge \text{NextCandidate}(i, \text{next}) \\ & \quad \wedge \forall q \in \text{IndexSet}(M) : \\ & \quad \quad \text{NextCandidate}(i, q) \\ & \quad \Rightarrow \\ & \quad e(\text{Delta}(M)[\text{next}]) \leq e(\text{Delta}(M)[q]) \end{aligned} $	$ \begin{aligned} & \text{next is an eligible successor} \\ & \text{next has minimal ending among eligible successors} \end{aligned} $
$ \begin{aligned} & \text{IsValid}(S) \triangleq \\ & \quad \wedge \text{FirstOK}(S[1]) \\ & \quad \wedge \forall r \in 1 \dots (\text{Len}(S) - 1) : \\ & \quad \quad \text{NextOK}(S[r], S[r + 1]) \\ & \quad \wedge \neg \exists q \in \text{IndexSet}(M) : \\ & \quad \quad \text{NextCandidate}(\text{Last}(S), q) \end{aligned} $	$ \begin{aligned} & \text{first index satisfies (3.1a)} \\ & \text{successive indices satisfy (3.1b)} \\ & \text{no successor exists after the last index: (3.1c)} \end{aligned} $
$ \begin{aligned} & \text{IN} \\ & \quad \text{CHOOSE } S \in \text{CandidateSequences} : \\ & \quad \quad \text{IsValid}(S) \end{aligned} $	$ \begin{aligned} & \text{choose one complete leading sequence} \end{aligned} $
$ \begin{aligned} & \text{IStar}(M) \triangleq \\ & \quad \text{I}^* = \{i_1, \dots, i_k\} \\ & \quad \text{Range}(\text{LeadingSequence}(M)) \end{aligned} $	$ \begin{aligned} & \text{image of the chosen leading sequence} \end{aligned} $
$ \begin{aligned} & \text{DeltaStar}(M, i) \triangleq \\ & \quad \text{IF } i \in \text{IStar}(M) \\ & \quad \quad \text{THEN } \text{SegTruncateLeft}(\text{Delta}(M)[i]) \\ & \quad \quad \text{ELSE } \text{Delta}(M)[i] \end{aligned} $	$ \begin{aligned} & \text{if } i \text{ is one of the leading indices} \\ & \quad \text{Delta_i}^* = - \text{Delta_i} \\ & \quad \text{otherwise } \text{Delta_i}^* = \text{Delta_i} \end{aligned} $
$ \begin{aligned} & \text{3.2} \\ & \text{MCross}(M) \triangleq \\ & \quad \text{LET} \\ & \quad \quad \text{KeptIndices} \triangleq \\ & \quad \quad \quad \{i \in \text{IndexSet}(M) : \\ & \quad \quad \quad \quad \text{DeltaStar}(M, i) \neq \text{EmptySeg}\} \\ & \quad \text{IN} \\ & \quad \quad \text{EnumerateValues}(\\ & \quad \quad \quad [i \in \text{KeptIndices} \mapsto \\ & \quad \quad \quad \quad \text{DeltaStar}(M, i)] \\ & \quad \quad) \end{aligned} $	$ \begin{aligned} & \text{discard empty transformed segments} \\ & \text{m}^{\wedge} \text{cross} = \text{sum of the nonempty } \text{Delta_i}^* \end{aligned} $
$ \text{DeltaCircle}(M) \triangleq $	

$Segment($
 $\quad min(M),$
 $\quad min(M) + Len(LeadingSequence(M)) - 1$
 $)$

left endpoint is $min\ m$
 right endpoint is $min\ m + k - 1$

$MW(M) \triangleq$
 $\langle MCross(M), DeltaCircle(M) \rangle$

$MW(m) = (m^{\wedge}cross, Delta^{\wedge}circ(m))$

Lemma 1

THEOREM $Lemma1 \triangleq$
 $\forall m \in (Multisegments \setminus \{\langle \rangle\}) :$
 $\quad min(m) < min(l(m))$
 $\Rightarrow$
 $\quad \wedge DerivedMultisegment(m) \neq \langle \rangle$
 $\quad \wedge MCross(m) \neq \langle \rangle$

PROOF OMITTED

The Statement - Corollary 3.4
 First alternative formulation

THEOREM $Corollary1 \triangleq$
 $\forall M \in (Multisegments \setminus \{\langle \rangle\}) :$
 $\quad min(M) < min(l(M))$
 $\Rightarrow$
 $\quad \wedge l(M) = l(MCross(M))$
 $\quad \wedge DeltaCircle(M)$
 $\quad \quad = DeltaCircle(DerivedMultisegment(M))$
 $\quad \wedge SameMultisegment($
 $\quad \quad \quad DerivedMultisegment(MCross(M)),$
 $\quad \quad \quad MCross(DerivedMultisegment(M))$
 $\quad \quad \quad)$

for every nonempty multisegment M
 assuming $min\ M < min\ l(M)$

$l(m) = l(m^{\wedge}cross)$

$Delta^{\wedge}circle(m) = Delta^{\wedge}circle(m')$

$(m^{\wedge}cross)' = (m')^{\wedge}cross$ as multisegments

PROOF OMITTED

The Statement - Corollary 3.4
 Second alternative formulation

THEOREM *Corollary2* $\triangleq$

LET

$KxId(P) \triangleq$

LET

$KP \triangleq K(P[1])$

IN

$\langle KP[1], KP[2], P[2] \rangle$

$IdxMW(P) \triangleq$

LET

$MWP \triangleq MW(P[2])$

IN

$\langle P[1], MWP[1], MWP[2] \rangle$

$SameResult(P, Q) \triangleq$

$\wedge P[1] = Q[1]$

$\wedge SameMultisegment(P[2], Q[2])$

$\wedge P[3] = Q[3]$

IN

$\forall M \in (Multisegments \setminus \{\langle \rangle\}) :$

$min(M) < min(l(M))$

$\Rightarrow$

$SameResult(
KxId(MW(M)),
IdxMW(K(M))
)$

apply K to the first component

$(K \times Id)(P)$

apply MW to the second component

$(Id \times MW)(P)$

ladder components are equal

multisegment components are equal

distinguished segments are equal

$(K \times Id)(MW(m)) = (Id \times MW)(K(m))$

PROOF OMITTED

The Statement - Corollary 3.4
Third alternative formulation

THEOREM *Corollary3* $\triangleq$

$\forall M \in (Multisegments \setminus \{\langle \rangle\}) :$

$min(M) < min(l(M))$

$\Rightarrow$

$\wedge K(M)[1] = K(MW(M)[1])[1]$

$\wedge MW(M)[2] = MW(K(M)[2])[2]$

$\wedge SameMultisegment(
K(MW(M)[1])[2],
MW(K(M)[2])[1]$

$)$

$l(m) = l(m^{\wedge cross})$

$Delta^{\wedge circ}(m) = Delta^{\wedge circ}(m')$

$(m^{\wedge cross})' = (m')^{\wedge cross}$ as multisegments

PROOF OMITTED
